# DF2 Vector surge and frequency protection relay

Table of content

1. Overview
2. Description
3. Function
   3.1 Vector surge relay
   3.2 Frequency
4. Settings
5. Technical data
6. Connection diagram




## 1. Overview

Combined frequency and vector surge relay serves to saving synchronous and asynchronous generators connected to the national grid. It has very high precision and speed due to microprocessor technology. but still has very simple settings.

## 2. Description

Measured AC signal is lead to input screws. It's shape is read by microprocessor. When input signal cross allowed limits, the appropriate output relay is open. Closed state of output relays is signaled by green LEDs.
Power supply is separate on the 230 V AC level.
Demanded parameters setting is possible by trimmers on the front side.

## 3. Functions

### 3.1 Vector surge relay

When the mains goes out, then generator works into too high or too low load depending on number of consumers on local net. By change of current in windings of generator the change of an angle between voltage and current vectors. It can be seen by measurement of voltage signal by indicating of voltage surge only in one single half-wave of voltage curve.
This safety relay counts the length of each half-wave, i,.e. each 10 milliseconds. When change of angle appropriate period length change is greater then which is set on left upper potentiometer, the output relay is opened for the time set by right upper potentiometer. Time from the mains failure to the disconnecting the output relay is 23 milliseconds maximally. (10 milliseconds to the end of half-wave and 13 ms for relay disconnecting)

### 3.2 Frequency relay

Each 10 milliseconds is measured the length of period of an input signal. The counted time is compared to the pre-set values of under and over frequency. When three following measurements are not within allowed limits, then output relay is opened. The time for letting the relay open is could be set by right upper potentiometer to the time from 2 to 10 milliseconds. It means that relay is again closed 2-10 seconds after restoration of good condition in the distribution network.
The under-frequency limit could be set by left lower potentiometer a over-frequency limit by right lower potentiometer.
Maximal time of opening of output contact is 53 milliseconds after overcrossing the allowed limits.

## 4. Settings

Parameters setting of relay is possible by four potentiometers and four DIL switches on the front side

| | | | |
|---|---|---|---|
| Switches: | 1 | ON | under-frequency relay activated |
| | 1 | OFF | under-frequency relay blocked / output relay always closed |
| | 2 | ON | over-frequency relay activated |
| | 2 | OFF | over-frequency relay blocked / output relay always closed |
| | 3 | ON | vector surge relay activated |
| | 3 | OFF | vector surge relay blocked / output relay always closed |
| | 4 | ON | Normal function of the whole relay |
| | 4 | OFF | internal test |

Internal test:

During internal test should be both power supply and measured signal connected to the national distribution net. the output relays are the close in order: 1st, 2nd, both and none. By this is tested good functionality of

- processor circuits
- measuring of an input signal
- closing of output relay

Potentiometers:

| | | | |
|---|---|---|---|
| Left upper | df | surge vector | 4 - 20 stupňů |
| Right upper | delay | time of restoration | 2 - 10 sec |
| Left lower | f < | under-frequency | 48 - 50 Hz |
| Right lower | f > | over-frequency | 50 - 52 Hz |

## 5. Technical data

| | |
|---|---|
| Dimensions HxWxD | 75x70x110 mm |
| Power supply | 230 V AC +- 20% |
| Power consuption | < 2 W |
| Internal fuse | 230V 50mA |
| Measured signal | 90 – 250V AC |
| | short time 300V |
| Input signal isolation | 5kV (opto-coupler) |
| Output relay contacts | Ag Cd O |
| Switched voltage max. | 250V AC / 125V DC |
| Switched power max. | 720 VA |
| Switched current max. | 6 A |
| Lifetime | 10 $10^6$ cycles |
| closing time | 11 ms |
| opening time | 13 ms |

## 6. Connection diagram

Output contact on diagram in position
"Operation without fault and good power supply"

